

ADVANCED SQL: NULL FUNCTIONS

EXERCISE 5

Table 1: Employees

employee-id	name	department	salary
1	Alice	HR	5000
2	Bob	IT	NULL
3	Charlie	NULL	7000
4	Dana	Finance	NULL

Q1. **SELECT** employee-id,
name,

IFNULL (salary, 0) **AS** salary-with-default
FROM Employees;

employee-id	name	salary
1	Alice	5000
2	Bob	0
3	Charlie	7000
4	Dana	0

Q2. **SELECT** employee-id,
name,

IFNULL (department, 'Not Assigned') **AS** department-name
FROM Employees;

employee-id	name	department-name
1	Alice	HR
2	Bob	IT
3	Charlie	Not Assigned
4	Dana	Finance

Table 2: Orders

order-id	customer-id	delivery-date
101	201	2024-12-01
102	202	NULL
103	NULL	2024-12-03

Q3. **SELECT** order-id,
customer-id
FROM Orders
WHERE customer-id IS NULL;

order-id	customer-id
103	NULL

Q4. **SELECT** *, IF NULL (delivery-date, 'Pending') AS delivery-status
FROM Orders;

order-id	customer-id	delivery-status
101	201	2024-12-01
102	202	Pending
103	NULL	2024-12-03

Table 3: Students

student-id	name	grade
1	Ethan	85
2	Maya	NULL
3	Olivia	90

Q5. **SELECT** student-id,
name,
IF NULL (grade, 0) AS final-grade $\Rightarrow$
FROM Students;

student-id	name	final-grade
1	Ethan	85
2	Maya	0
3	Olivia	90

Q6. **SELECT** COUNT (*) AS not-graded-count
FROM Students
WHERE grade IS NULL;

not-graded-count
1

Table 4: Products

product-id	name	price	discount
501	Keyboard	25	NULL
502	Mouse	15	5
503	Monitor	100	NULL

Q7. **SELECT** product-id,
name,
price,
IF NULL (discount, 0) AS final-price
FROM Products;

product-id	name	price	final-price
501	Keyboard	25	0
502	Mouse	15	5
503	Monitor	100	0

Table 5 : Customers

customer_id	name	email
1	Linda	NULL
2	Joseph	joseph@gmail.com
3	Nia	NULL

Q8. **SELECT COUNT(*) AS missing_email_count**
FROM Customers
WHERE email IS NULL ;

missing_email_count
2

Q9. **SELECT customer_id,**
name,
IF NULL (email, 'No Email') AS email_display
FROM Customers ;

customer_id	name	email_display
1	Linda	No Email
2	Joseph	joseph@gmail.com
3	Nia	No Email

Table 6: Payments

payment-id	method	status
301	Credit	NULL
302	PayPal	Success
303	NULL	Failed

Q10. **SELECT** payment-id,
 IF NULL (method, 'Unknown') AS method-display,
 status
 FROM Payments;

payment-id	method-display	status
301	Credit	NULL
302	PayPal	Success
303	Unknown	Failed

Table 7: Inventory

item-id	item-name	quantity
1	Pen	NULL
2	Notebook	150
3	Eraser	NULL

Q11. **SELECT** item-id,
 item-name,
 IF NULL (quantity, 0) AS quantity-checked
 FROM Inventory;

item-id	item-name	quantity-checked
1	Pen	0
2	Notebook	150
3	Eraser	0

Table 8: Employees-Extra

emp-id	bonus	commission
1	NULL	300
2	100	NULL
3	NULL	NULL

Q12. SELECT emp-id,

COALESCE (bonus, commission) AS first-available-reward

FROM Employees-Extra;

emp-id	first-available-reward
1	300
2	100
3	NULL

Table 9: Classes

class-id	subject	room
11	Math	NULL
12	Science	Lab A
13	English	NULL

Q13. SELECT COUNT (*) AS no-room-count

FROM classes

WHERE room IS NULL;

no-room-count
2

Table 10 : Attendance

student-id	date	status
1	2025-04-01	NULL
2	2025-04-01	Present
3	2025-04-01	Absent

Q14. **SELECT** student-id,
date,

IF NULL (status, 'Not Marked') **AS** attendance-status
FROM Attendance ;

student-id	date	attendance-status
1	2025-04-01	Not Marked
2	2025-04-01	Present
3	2025-04-01	Absent

Table 11 : Bank-Accounts

account-id	account-type	balance
A1	Savings	NULL
A2	Current	5000
A3	NULL	2000

Q15. **SELECT** account-id,
COALESCE (account-type, 'Unknown') **AS** type-display,
COALESCE (balance, 0) **AS** balance-checked
FROM Bank-Accounts ;

account-id	type-display	balance
A1	Savings	0
A2	Current	5000
A3	Unknown	2000

Table 12: Projects

project-id	title	start-date	end-date
1	Website Revamp	2025-01-10	NULL
2	Mobile App	NULL	2025-06-01
3	Data Migration	NULL	NULL

Q16. **SELECT** project-id,
title;

IF NULL (start-date, 'TBD') **AS** start-display
FROM Projects;

project-id	title	start-display
1	Website Revamp	2025-01-10
2	Mobile App	TBD
3	Data Migration	TBD

Table 13: Reviews

review-id	product-id	comment	rating
1	501	Great product	4
2	502	NULL	NULL
3	503	Works fine	3

Q17. **SELECT** review-id,
product-id;

IF NULL (comment, 'No Comment') **AS** comment-display,

IF NULL (rating, 0) **AS** rating-display

FROM Reviews;

review-id	product-id	comment-display	rating-display
1	501	Great product	4
2	502	No Comment	0
3	503	Works fine	3

Table 14: Suppliers

supplier-id	name	phone	alt-phone
1	Global Goods	NULL	123456789
2	Best Supplies	987654321	NULL
3	Value Source	NULL	NULL

Q18. `SELECT supplier-id,
name,
COALESCE(phone, alt-phone, 'No Contact') AS contact-number
FROM Suppliers;`

supplier-id	name	contact-number
1	Global Goods	123456789
2	Best Supplies	987654321
3	Value Source	No Contact

Table 15: User-Settings

user-id	theme	language	time zone
1	NULL	English	NULL
2	Dark	NULL	UTC+1
3	NULL	NULL	NULL

Q19. `SELECT user-id,
theme,
IF NULL (theme, 'Light') AS theme-set,
IF NULL (language, 'English') AS language-set,
IF NULL (time zone, 'UTC') AS time zone-set
FROM User-Settings;`

user-id	theme-set	language-set	time zone-set
1	Light	English	UTC
2	Dark	English	UTC +1
3	Light	English	UTC

Table 16: Maintenance

record-id	machine-id	issue	technician
1	M101	Overheating	NULL
2	M102	NULL	NULL
3	M103	Jammed	Alex

Q 20. `SELECT record-id,
machine-id,
IF NULL (issue, 'Unknown Issue') AS issue-log,
IF NULL (technician, 'Not Assigned') AS technician-name
FROM Maintenance;`

record-id	machine-id	issue-log	technician-name
1	M101	Overheating	Not Assigned
2	M102	Unknown Issue	Not Assigned
3	M103	Jammed	Alex